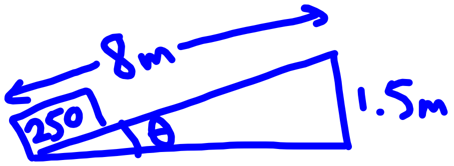
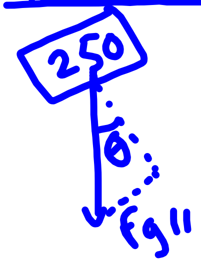
Energy Unit Review

1.



a) $W = \Delta E_p = mgh - 0$
$= 250(9.8)1.5$
$\boxed{= 3680\,J \text{ or } 3.68\,kJ}$

b) force components:



$F_{g\parallel} = mg\sin\theta$
$= 250(9.8)\left(\frac{1.5}{8}\right)$
$\boxed{= 459\,N}$

work:

$W = F \cdot d$

$3680 = F \cdot 8$

$\boxed{F = 459\,N}$

---

2. a) $W = \Delta E_p = mgh_f - mgh_i = mg(h_f - h_i)$

$W = 3.5(9.8)(2.5 - 1.45) = \boxed{36\,J}$

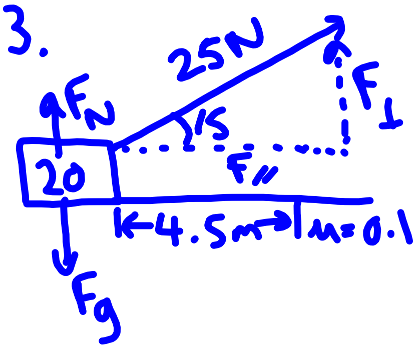
---

b) $W + \cancel{E_{pi}} + \cancel{E_{ki}} = E_{pf} + E_{kf}$

$F \cdot d = mgh_f + \frac{1}{2}mv_f^2$

$1100(12) = 200(9.8)(6) + \frac{1}{2}(200)v_f^2 \Rightarrow \boxed{v_f = 3.8\,m/s}$

3.



$F_N + F_\perp = F_g$

$F_N = F_g - F_N$

a) $W_p = F_\parallel \cdot d = 25\cos 15 \cdot 4.5 = \boxed{109\,J}$

b) $W_f = \overset{\mu F_N}{F_f} \cdot d = \mu(F_g - F_\perp) \cdot d$
$= 0.1[20(9.8) - 25\sin 15] \cdot 4.5$
$= \boxed{85.3\,J}$

c) $W_p = W_f + \Delta E_k$

$109 = 85.3 + \left(\frac{1}{2}(20)v^2 - 0\right)$

$\boxed{v = 1.5\,m/s}$

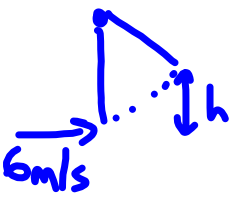
4. W = area under graph:

a) $\triangle\ \frac{1}{2}(0.02-0)(60) + \square(0.04-0.02)(60) = \boxed{1.8\text{ J}}$

b) $\triangle\ \frac{1}{2}(0.02-0)(60) + \square(0.06-0.02)(60) + \triangle\ \frac{1}{2}(0.1-0.06)(60)$
$= \boxed{4.2\text{ J}}$

c) $\square(0.06-0.04)(60) + \triangle\ \frac{1}{2}(0.1-0.06)(60) = \boxed{2.4\text{ J}}$

5. a)



$E_k = E_p$

$\frac{1}{2}\cancel{m}v^2 = \cancel{m}gh$

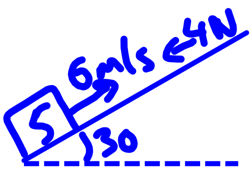
$\frac{1}{2}(6)^2 = (9.8)h$

$\boxed{h = 1.84\text{ m}}$

b) $E_{pi} + \cancel{E_{ki}} = \cancel{E_{pf}} + E_{kf} \Rightarrow \cancel{m}gh_i = \frac{1}{2}\cancel{m}v_f^2$

$9.8(4.20) = \frac{1}{2}v_f^2 \Rightarrow \boxed{v_f = 9.07\text{ m/s}}$

c)

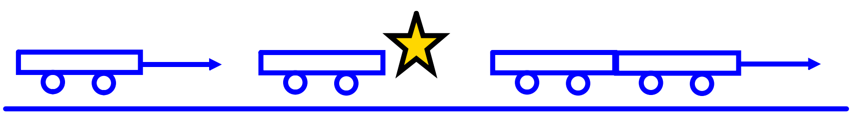


$\cancel{W} + E_{ki} + \cancel{E_{pi}} = \cancel{E_{kf}} + E_{pf} + E_{Loss}$ ← $F_f \cdot d$

E_{pf} ← $mg(\sin\theta \cdot d)$

$\frac{1}{2}(5)(6)^2 = (5(9.8)\sin 30 + 4)\cdot d \Rightarrow \boxed{d = 3.2\text{ m}}$

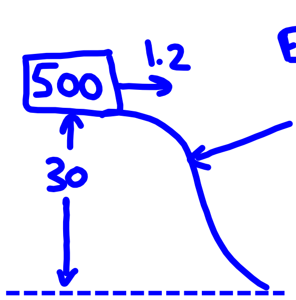
6.



$E_{ki} = E_{kf} \Rightarrow \frac{1}{2}(8)(8)^2 = \frac{1}{2}(8+5)v_f^2$

$\boxed{v_f = 6.3\text{ m/s}}$

7. a)



$E_T = E_k + E_p = \frac{1}{2}mv^2 + mgh = \frac{1}{2}(500)(1.2)^2 + 500(9.8)(30)$
$= 147360$

i) $\frac{1}{2}(500)v^2 + 500(9.8)(25) = 147360 \Rightarrow \boxed{v = 10\,m/s}$

ii) $\frac{1}{2}(500)v^2 + 500(9.8)(12) = 147360 \Rightarrow \boxed{v = 18.8\,m/s}$

iii) $\frac{1}{2}(500)v^2 + 500(9.8)(0) = 147360 \Rightarrow \boxed{v = 24.3\,m/s}$

---

8. a) i) $\frac{1}{2}\cancel{m}v_i^2 + \cancel{m}gh_i = \frac{1}{2}\cancel{m}v_f^2 + \cancel{m}gh_f$

$9.8(12) = \frac{1}{2}v_f^2 + 9.8(4)$

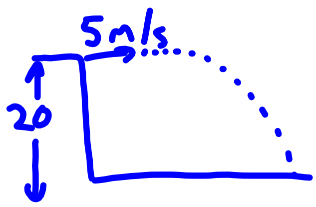
$\boxed{v_f = 13\,m/s}$

ii) $9.8(12) = \frac{1}{2}v_f^2 + 9.8(2)$

$\boxed{v_f = 14\,m/s}$

---

b)



$E_k + E_p = E_k + 0$

$\frac{1}{2}\cancel{m}v_i^2 + \cancel{m}gh = \frac{1}{2}\cancel{m}v_f^2$

$\frac{1}{2}(5)^2 + 9.8(20) = \frac{1}{2}v_f^2$

$\boxed{v_f = 20.4\,m/s}$

---

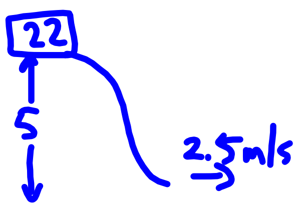
9. a) $\cancel{\frac{1}{2}mv_i^2} + mgh_i = \frac{1}{2}mv_f^2 + \cancel{mgh_f} + E_{Loss}$

$1.0(9.8)(2) = \frac{1}{2}(1.0)(5.3)^2 + E_{Loss}$

$\boxed{E_{Loss} = 5.6\,J}$

---

b)



$\cancel{\frac{1}{2}m0^2} + mgh = \frac{1}{2}mv^2 + \cancel{mg(0)} + E_{Loss}$

$22(9.8)(5) = \frac{1}{2}(22)(2.5)^2 + E_{Loss}$

$\boxed{E_{Loss} = 1010\,J \text{ or } 1.01\,kJ}$

10.

$F_f = \mu mg$

$P = \dfrac{W}{t} = \dfrac{F_f \cdot d}{t} = \dfrac{0.02(30)(9.8) \cdot 10}{20}$

$\boxed{P = 2.94\,W}$

---

11. a)

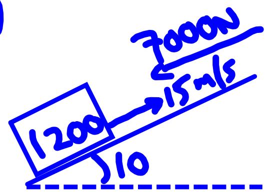
i) $F = mg\sin\theta = 6(9.8)\sin(8.75) = \boxed{8.9\,N}$

ii) $W = F \cdot d = 8.9(10) = \boxed{89\,J}$

iii) $P = F \cdot v = 8.9(1.5) = \boxed{13\,W}$

- -

b)



i) F = overcome gravity & drag

$= mg\sin\theta + F_{drag}$

$= 1200(9.8)\sin 10 + 7000$

$\boxed{F = 9000N}$

ii) $P = F \cdot v = 9000 \cdot 15 = \boxed{136\,kW}$

$136\,kJ \cdot \dfrac{1000\,J}{kJ} \cdot \dfrac{1\,hp}{750\,W} = \boxed{181\,hp}$

---

12. a) $eff = \dfrac{W_{out}}{W_{in}} = \dfrac{\Delta E_p}{W_{in}} = \dfrac{mg(h_f - h_i)}{W_{in}} = \dfrac{80(9.8)(24-18)}{18000}$

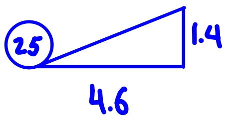
$= \boxed{26\%}$

- -

b) $eff = \dfrac{P_{out}}{P_{in}} = \dfrac{\left(\frac{W}{t}\right)}{P_{rating}} = \dfrac{\left[\frac{(mgh - 0)}{t}\right]}{P_{rating}}$

$0.82 = \dfrac{\left(\dfrac{1200(9.8)(30)}{1.5\,\cancel{min} \cdot \frac{60\,s}{\cancel{min}}}\right)}{P_{rating}} \Rightarrow \boxed{P_{rating} = 4.8\,kW}$

13.



a) $\underset{\text{(ramp)}}{\text{IMA}} = \dfrac{\text{slope}}{\text{rise}} = \dfrac{\sqrt{1.4^2+4.6^2}}{1.4}$ (pythag) $= \boxed{3.4}$

b) $\text{AMA} = \dfrac{\text{load}}{\text{effort}} = \dfrac{25(9.8)}{95} = \boxed{2.6}$ (mg)

c) $\text{eff} = \dfrac{\text{AMA}}{\text{IMA}} = \dfrac{2.6}{3.4} = \boxed{75\%}$

$\text{eff} = \dfrac{W_{out}}{W_{in}} = \dfrac{25(9.8)1.4}{95\sqrt{1.4^2+4.6^2}} = \boxed{75\%}$

---

14. a) $\underset{\text{(lever)}}{\text{IMA}} = \dfrac{\text{effort arm}}{\text{load arm}} = \dfrac{2.1}{0.265} = \boxed{7.9}$

b) $\text{AMA} = \dfrac{\text{load}}{\text{effort}} = \dfrac{68(9.8)}{150} = \boxed{4.4}$

c) $\text{eff} = \dfrac{\text{AMA}}{\text{IMA}} = \boxed{56\%}$

---

15. a) $\text{AMA} = \dfrac{640}{75} = 8.5$

b) $\text{eff} = \dfrac{\text{AMA}}{\text{IMA}} = \dfrac{\text{AMA}}{\left(\dfrac{\text{length pulled}}{\text{distance moved}}\right)} \Rightarrow 0.85 = \dfrac{8.5}{\left(\dfrac{\ell}{0.21}\right)}$

$\boxed{\ell = 2.1\text{ m} = 210\text{ cm}}$

---

16. a) $\text{eff} = \dfrac{W_{out}}{W_{in}} = \dfrac{mgh - 0}{F\underbrace{[(2\pi r)\cdot\text{revolutions}]}_{d}}$

$= \dfrac{15(9.8)(45)}{11.2(2\pi\left(\frac{38}{2}\right)(70)} = \boxed{71\%}$

---

b)

$\text{eff} = \dfrac{\text{AMA}}{\text{IMA}} = \dfrac{\left(\dfrac{\text{load}}{\text{effort}}\right)}{\left(\dfrac{\text{wheel diameter}}{\text{axle diameter}}\right)}$

$0.71 = \dfrac{\left(\dfrac{15(9.8)}{11.2}\right)}{\left(\dfrac{38}{\text{axle}}\right)} \Rightarrow \dfrac{0.71(38)}{\text{axle}} = \dfrac{15(9.8)}{11.2}$

$\text{axle} = \boxed{2.0\text{ cm}}$